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Short-term multi-hour ahead country-wide wind power prediction for Germany using gated recurrent unit deep learning

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ABSTRACT

In recent years, wind power has emerged as an important source of renewable energy. When onshore and offshore wind farm regions are connected to the grid for power generation, consistent multi-location short-term wind power predictions are extremely valuable in terms of assuring the power system's safety, sustainability, and economic operation. An abrupt variation in wind power generation influences the efficiency of the regional power grid. This makes accurate short-term forecasting essential for high-level planning and scheduling of power grids. To address the issue, this paper presents two variants of recurrent neural networks (RNN): gated recurrent unit (GRU) and long short-term memory (LSTM) models considering substantially better prediction accuracy to forecast a country-wide (Germany) wind power data for daily ($t+1$), and multi-step ($t+3$, $t+5$, and $t+12$) hours ahead. In addition, wind velocities [m/s] measured at heights of 2, 10, and 50-m (above ground level) are exploited as an essential characteristic among the available input variables and evaluated each feature subset based on four training divisions (80-20%, 70-30%, 60-40%, and 50-50%) and compared the results with ARIMA and SVR approaches in the literature. The findings reveal that the RNN-GRU model not only can achieve higher predicting accuracy but also has a faster learning speed over long sequences.

1. Introduction

The continuous rise in global energy use, and a widespread desire to minimize the reliance on fossil fuels due to their negative environmental consequences, have led to the significant demand and development of renewable power sources, particularly solar and wind. However, Wind intermittency and short-term unpredictability bring various challenges to forecasting, planning, and effectively integrating its supply into power systems to meet demand changes. Plant operators, grid system controllers, regulators, and energy market planners all seek to obtain accurate wind power forecasts over the short and medium term; this helps them to stabilize the power supply and assists them in scheduling standby supplies and managing the supply to meet peak power demand periods. Therefore, short-term wind power forecasts over the next few hours are essential at the local level for individual plants and the regional and national level for system controllers and planners.

Wind power generation has become a key supply component in many contemporary power systems because of its cost-competitiveness and its valuable contribution to reducing greenhouse gas emissions. According to the International Renewable Energy Agency, the global installed capacity of wind power reached 733.3 GW in 2020, up from 416.3 GW in 2015 [1]. Indeed, the global expansion of wind power is almost doubled every year. Twelve countries improved their wind power capacity by over 1 GW in 2020. [2], specifically, Germany increased its wind power capacity from 44.6 GW in 2015 to 62.2 GW in 2020. Moreover, offshore wind installations are an important component, which is now being installed to further increase the country's capacity by about 20 GW before 2030 [3]. Large-scale wind power integration commenced in Germany to overcome the problems associated with its highly distributed nature as well as its difficult-to-predict intermittency and potential instability [4]. Additionally, the quantity of wind energy generated at a specific location is proportional to the wind speed unit, which indicates that a 10% increase in wind speed leads to a 30% increase in wind

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Nomenclature

RMSE	root mean square error
GRU	Gated recurrent unit
RNN	Recurrent neural network
$\otimes$	Point-wise multiplication
DL	Deep learning
w_{hf}	Weight of forget gate
$\Re$	Number of samples/datasets
ψ	*Tanh*
LSTM	Long short-term memory
RBF	Radial basis function
x, y	Represents the actual and desired values
ARIMA	Autoregressive integrated moving average
NN	neural network
h	The activation function in GRU cell
CEC	Constant error carousel
SVR	support vector regression
OPSD	Open Power System Data
$y_{(actual)}$	Actual value
MAE	Mean absolute error
σ	Sigmoid
b_o	Bias value for output gate
$\widetilde{h}$	The candidate activation function in GRU cell

energy output. As a result, different wind power forecasting models depend upon accurate recoding of wind speed at relevant elevations above ground level as a significant input variable measurement error in wind speed is one of the major causes of error in wind power prediction [5]. Despite this, wind speed at most locations is strongly nonlinear and unpredictable in behavior with irregular patterns, and wind power is influenced by several variables such as; elevation, climate, air pressure, temperature, and surface roughness for wind forecasting [6]. Therefore, accurate wind power prediction (WPP) is an indispensable aspect of wind farm grid connection; in particular, the ultra-short-term WPP of wind farms can effectively guide power generation, dispatch, operation, and maintenance.

Many WPP methods have been developed, which can be broadly divided into two paradigms: physical methods and statistical methods. To configure wind farms, physical approaches consider a variety of variables, mostly environmental, geographical, and meteorological, and depend on numerical weather prediction (NWP) tools that continually record local atmospheric conditions. Additionally, Statistical models are performed by establishing relationships among wind power and variables affecting wind power based on historical data [7–9]. Both physical and statistical models require long-term operational records and sufficient computational resources to be effective and accurate. Models aimed at improving wind power forecast accuracy are necessary since they help to improve the stability of the grid and also reduce the grid operational costs [10,11]. Furthermore, statistical methods include regression analysis [12], exponential smoothing [13], and applying a Kalman filter [14]. The fundamental idea is to apply the statistical laws governing previous and current WP data to predict future power values. The disadvantage of statistical methods lies in their limited ability to predict nonlinear relational data. Shallow machine learning methods, such as extreme learning machines [15], random forests [16], and support vector regression (SVR) [17]generally require complex feature extraction and enhancement to supplement temporal features to more effectively understand the correlations among time series data and ensure prediction accuracy.

For time series analysis, many regression-based approaches have been proposed to address the volatility of large-scale wind power supply, fit the fluctuation of wind speed, and quantify the uncertainty associated with multiple hour-ahead forecasts [18], such as quartile regression [19,20], ensemble methods, adaptive resampling [21,22], radial basis function (RBF) neural network [23,24], bootstrapped extreme learning machine (ELM) [25], conditional kernel density estimation [26], optimized data matching [27], and direct interval forecasting. All these methods aim to potentially boost hour-ahead prediction precision. Yuan *et* al employed an autoregressive fractionally integrated moving average model and the least square support vector machine model to forecast the linear and nonlinear components of wind power, respectively [28]. Tian proposed a local mean decomposition to divide the short-term wind speed time series into several components, which were then predicted using a combined kernel function least squares support vector machine [29]. For wind speed prediction, the bagging ensembles of decision trees approach was used to take advantage of many relatively weak single trees [30]. However, complex nonlinear features of very large wind data cannot be easily and rapidly captured shallow machine learning models. Consequently, such models tend to generate sub-optional hour-ahead prediction.

The majority of popular artificial intelligence-based forecasting algorithms are focused on single forecast models. However, finding an individual forecast model that is ideal for all types of wind power supply is difficult. This is particularly true for probabilistic models based on probability distribution assumptions applied to historic data since wind generation characteristics vary significantly from one location and region to another. Zuluaga *et* al. proposed a Kalman filter-based wind speed forecasting model by using the previous hourly average wind speed, which was built for one-day-ahead wind speed prediction. Another study suggested that wind speed can be predicted using a combination of empirical mode decomposition (EMD) and kernel ridge regression [31]. Hu *et*.al proposed a prediction model based on empirical wavelet transform, partial autocorrelation function, and Gaussian process regression for short-term wind speed prediction [32]. A prediction interval model was built by combining variation mode decomposition (VMD) and low-rank multi-kernel ridge regression. In Ref. [33], VMD has also been employed to decompose wind speed into the nonlinear part, the linear part, and the noise components, and the autoregressive and moving average models under the Monte Carlo Markov Chain framework were then introduced for the prediction of the linear component. Although such models are relatively easy to apply to time series analysis, they are only suitable for predicting data with linear characteristics. Previous research has demonstrated that aggregating the mean of predictions from several models typically provides more reliable prediction performance than that achieved by individual models. Of different models outperforms each model [34]. Ensemble probabilistic wind power forecasting integrates various weather-related variables, including wind speed and rainfall, by applying Bayesian model averaging (BMA) [35]which combines distributions of the same type.

An intelligent ensemble regressor that combined artificial neural networks and genetic programming was constructed for the forecast of wind power [36]. In Ref. [37], The ensemble empirical mode decomposition approach has also divided the original wind speed data into multiple signals. These were then regressed to predict future wind speed values using the adaptive wavelet neural network model. Wang *et*.al suggested a model that combines an extreme learning machine with an improved complementary ensemble empirical mode decomposition with adaptive noise (ICEEMDAN) and an autoregressive integrated moving average (ARIMA) [38].

Additionally, machine learning models, including artificial neural networks (ANN) [39], graph convolutional networks (GCN) [40], and stacked denoising auto-encoder (SDAE) also make persistence

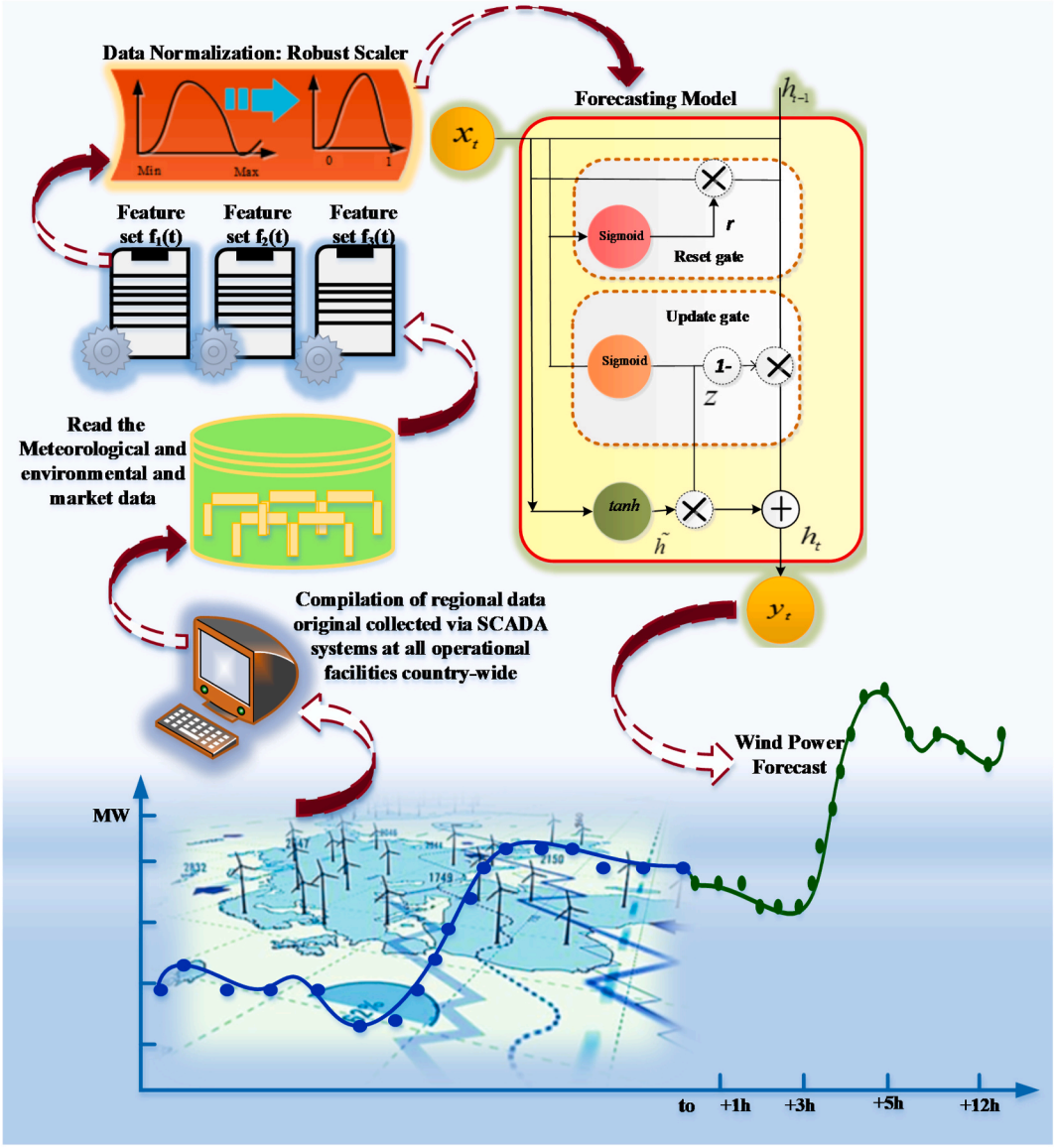


Fig. 1. Diagrammatic illustration of the wind power forecasting sequence for the RNN models.

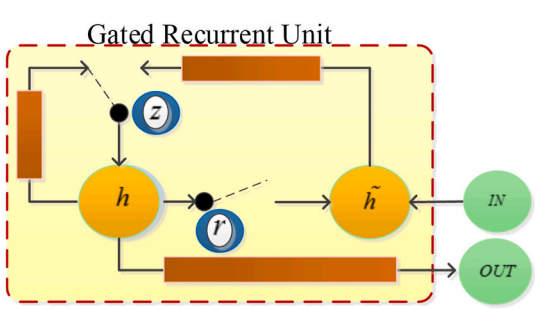


Fig. 2. Basic configuration of a GRU; z and r refer to the update and reset gates, respectively, whereas h and $\tilde{h}$ refer to the activation function and the candidate activation function, respectively.

assumptions [41]. Decomposing wind power time series using various wavelet decomposition (WD) [42] techniques can reduce the volatility of the signal by dividing it into simple parts, making it easier for prediction models to process. Recurrent neural networks (RNN), such as the long short-term memory (LSTM) with wavelet activation functions focus on capturing periodic dependencies over time intervals, making them ideal for wind power prediction. Altan *et* al. introduced ensemble empirical mode decomposition (EEMD), a method for analyzing noise data that decomposes time series into distinct frequency sections for training the LSTM model to increase the prediction accuracy of crude oil prices [43]. Karasu *et* al.*et* al developed a hybrid WSF model based on LSTM network and decomposition methods combined with a grey wolf optimizer (GWO) to deal with the missing data by the weighted moving average (WMA) method. The WS time series (WSTS) data are then smoothed by WMA filtering to forecast wind power [44].

Furthermore, as a LSTM variant, the gated recurrent unit (GRU) network has a simplified configuration that involves fewer training parameters. In several application areas, it has been demonstrated an ability to accomplish similar prediction accuracy to LSTM while involving fewer iterations [45]. GRU and LSTM models have the advantage over standard ANN models in that they can hold lag periods of historical time series data in memory to enhance multistep ahead prediction capabilities. Variants of recurrent neural networks, including long short-term memory (LSTM), and gated recurrent units (GRUs), have been used for WPP to eliminate the complexity involved with feature augmentation [46]. Because these approaches utilize deep

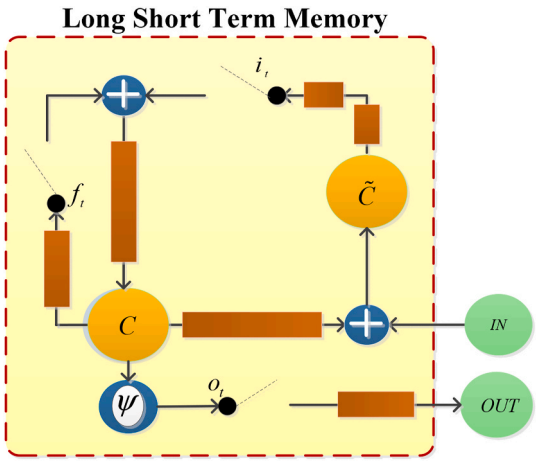


Fig. 3. Basic configuration of an LSTM model. The black circles depict the three-component states: input i_t, output o_t, and forget gate f_t, and the current memory cell state C and new candidate values for cell $\widetilde{C}$ along with the ψ (tanh activation function) are learned by considering information at the previous, current, and next time steps, that is $t-1$, t and $t+1$.

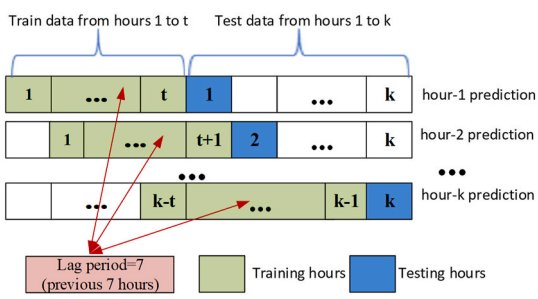


Fig. 4. Learning principle of time steps in RNN-GRU.

Table 1
Value selection for the lag (L) parameter based on trial and error assessments of f_1 (t) feature set for Case 4.

L	MAE	RMSE	R^2
3	40.72	75.65	0.97
5	41.49	76.21	0.98
7	24.23	53.08	0.99

network architectures in general, they have excellent nonlinear adaptation capabilities and can learn the temporal correlation properties of time series data effectively. These types of recurrent neural network models were only used for single-step predictions in the above-mentioned reports; while this approach can achieve high prediction accuracy by evaluating information from points near that to be predicted, it cannot guarantee accurate multi-step prediction results. In particular, the conventional approaches have shortcomings i. e, the minimal computational efficiency in terms of spatial correlation modeling and the inadequate learning ability of RNN models in terms of time-correlation modeling. Considering the analysis presented above, the multi-location multi-step WPP model is based on the RNN time series regression method. Furthermore, the proposed model has an excellent nonlinearity approximation ability and better generalization capability. Real wind power data collected from an offshore wind farm in Germany is used to verify the effectiveness of the proposed model.

This study describes the systematic structure for a cascaded wind-power forecasting model employing the RNN-GRU and LSTM algorithms. Section 2 describes the generic framework required for multi-step ahead wind power forecast models. Section 3 details the proposed combined LSTM/GRU model evaluated. Section 4 describes the German country-wide wind power dataset for 2016 evaluated for forecasting. Section 5 examines the comprehensive forecasting outcomes of the RNN techniques applied and compares their results with ARIMA and SVR models. Section 6 discusses possible applications and future developments for the models evaluated.

Table 2
Configuration and control parameter settings applied to SVR (RBF), RNN-GRU, and RNN-LSTM Models.

Model	Parameters	Value
SVR	Kernel	RBF
	Degree of polynomial	3
	C	100
	Epsilon	0.0000001
RNN-GRU/RNN-LSTM	Activation function	Tanh
	Layer	2
	Dropout	0.2
	No. Of neurons	{128, 80, 64, 32, 16}
	Learning rate	0.001
	Optimizer	Adam
	Batch size	32
	Epochs	300

2. Material and methods

2.1. RNN models developed for wind power prediction

In this work, two types of deep learning-based time series regression models are developed for wind power forecasts compiling hourly data records for a dataset integrating country-wide regions of Germany (see Section 3). Variants of RNN models are developed, i.e., RNN-GRU and RNN-LSTM. A general workflow schematic (Fig. 1) illustrates the essential steps and their application sequence for the RNN models.

2.1.1. Gated recurrent unit

The gated recurrent unit (GRU) consists of two gates: a reset gate, r_t and an update gate, z_t [47]. The $\widetilde{h}_t$ term in Equation 3 represents the memory unit enabling the reset gate to hold and exploit some of the previously processed information. Neither of the mentioned gates possesses its separate memory units [48]. The update gate is employed when an existing hidden state requires updating with a new hidden state. On the other hand, the reset gate is employed to ignore, purge or

Table 3
The training and testing subset sizes evaluated as four distinct cases.

	Training subset size (samples)	Testing subset size (samples)
Case 1 (80-20%)	7028	1756
Case 2 (70-30%)	6148	2636
Case 3 (60-40%)	3514	5270
Case 4 (50-50%)	4392	4392

Table 4
Statistical description of features for the Germany wind power dataset.

	Min	Max	Mean
Seq#	1	8784	4392.5
V1 (m/s)	0.7465	7.9277	2.9448
V10 (m/s)	1.3455	11.1643	4.2978
V50 (m/s)	1.9590	14.9230	6.0579
Z0(m)	0.1409	0.2352	0.2003
T (°C)	−8.5531	28.6953	9.2220
Rho (kg/m^3)	1.1343	1.3069	1.2117
Pa (pa)	95467.6	100956.5	98551.4
PrI (Euro/MW)	−5.4886	2.6215	0.0000
MW	135	33,626	8766.8

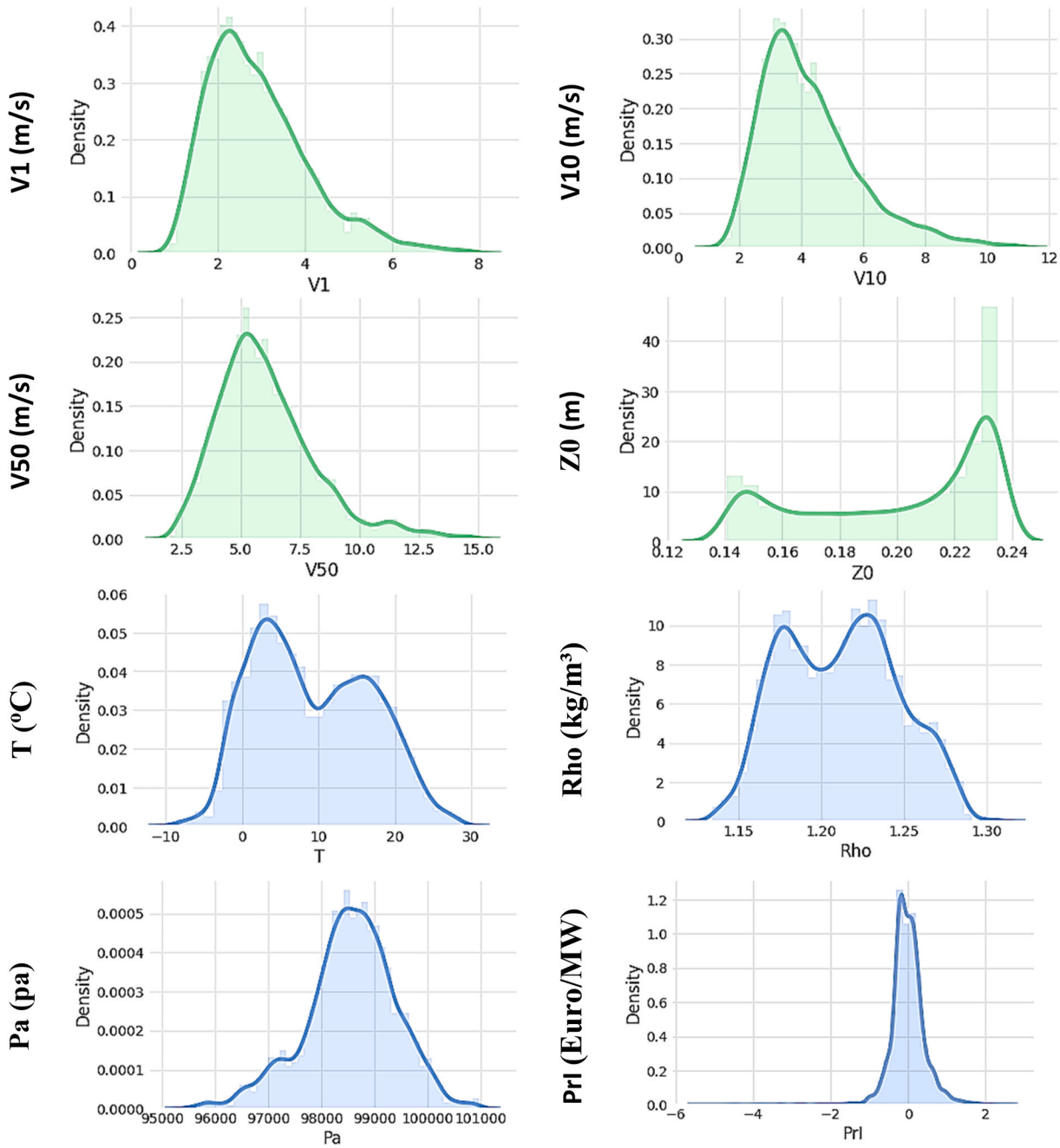


Fig. 5. Value distributions for each feature in the evaluated wind power dataset.

override prior hidden states. Fig. 2 provides a diagrammatic representation of the GRU cell. The reset gate works analogously to the forget gate of the LSTM (see section 2.2.2). A hidden unit at the time step t is computed by taking into account the hidden unit at a time step $t-1$. The reset gate, update gate, and hidden unit at a time t are represented by equations (1)–(3).

$$z_t = \sigma(w_z x_t, w_{hz} h_{t-1} + b_z) \tag{1}$$

$$r_t = \sigma(w_r x_t, w_{hr} h_{t-1} + b_r) \tag{2}$$

$$\begin{aligned} \widetilde{h}_t &= \tanh(w_h \cdot [r_t \otimes h_{t-1}, x_t] + b_h) \\ h_t &= (1 - z_t) \otimes h_{t-1} + z_t \otimes \widetilde{h}_t \end{aligned} \tag{3}$$

2.1.2. Long short-term memory

LSTM is a form of RNN that is initiated by changing the structure of its recurrent connections [49]. As a self-connected unit, the LSTM acts essentially as a constant error carousel (CEC) that considers and adapts to the flow of errors over time. It incorporates input, output, and forget gates to maintain and adapt to the content of the memory stored as part of its CEC function. This protects the other components of the LSTM from unrelated memory contents. It also enables the CEC memory to be periodically reconstructed once its contents become ineffective and superseded by updated information. Fig. 3 is a diagrammatic illustration of a signal LSTM cell.

The architecture of the LSTM model comprises three gates named input, forget, and output gates by applying modulation functions to the input and output. The information at the gates typically involves current and recurrent inputs adapted through sigmoid functions (σ). On the other hand, the modulation process usually applies a hyperbolic tangent activation function ($tanh\ \psi$).

Fig. 6. The correlation coefficient matrix heat map for the wind power variables considering values from all 8784 data records of the evaluated dataset.

Table 5
Performance analysis of proposed scheme (RNN-GRU, and RNN-LSTM) on test data in terms of all cases (80-20%, 70-30%, 60-40%, and 50-50% division of dataset) for feature sets (f_1 (t), f_2 (t), and f_3 (t)).

Feature-set	Scenarios	Models	MAE	RMSE	R^2
$f_1(t)$	Case 1	RNN-GRU	**44.017**	**97.69**	**0.99**
		RNN-LSTM	45.06	95.73	0.99
	Case 2	RNN-GRU	**26.45**	**73.01**	**0.99**
		RNN-LSTM	43.23	112.53	0.99
	Case 3	RNN-GRU	**30.40**	**55.61**	**0.99**
		RNN-LSTM	40.83	86.72	0.99
	Case 4	RNN-GRU	**24.23**	**53.08**	**0.99**
		RNN-LSTM	42.21	84.08	0.99
$f_2(t)$	Case 1	RNN-GRU	**95.07**	**186.41**	**0.99**
		RNN-LSTM	155.37	244.62	0.99
	Case 2	RNN-GRU	**106.76**	**231.55**	**0.99**
		RNN-LSTM	129.87	234.45	0.99
	Case 3	RNN-GRU	**99.03**	**202.81**	**0.99**
		RNN-LSTM	126.37	215.99	0.99
	Case 4	RNN-GRU	**95.44**	**195.91**	**0.99**
		RNN-LSTM	154.49	221.11	0.99
$f_3(t)$	Case 1	RNN-GRU	**97.71**	**187.61**	**0.99**
		RNN-LSTM	108.90	209.84	0.99
	Case 2	RNN-GRU	**67.24**	**125.08**	**0.99**
		RNN-LSTM	71.11	145.63	0.99
	Case 3	RNN-GRU	**76.75**	**153.60**	**0.99**
		RNN-LSTM	88.22	167.24	0.99
	Case 4	RNN-GRU	**127.61**	**203.20**	**0.99**
		RNN-LSTM	161.02	261.38	0.99

However, other activation functions such as "ReLU" and "leaky ReLU" can be used and it is generally worthwhile to conduct sensitivity analysis to establish which specific modulation/activation function works best with specific data sets. Therefore, to capture possible long-term dependencies, each hidden state has to compute equations (4)–(8) at each time step (t).

$$f_t = \sigma\left(w_f x_t + w_{hf} h_{t-1} + b_f\right) \tag{4}$$

$$i_t = \sigma(w_i x_t + w_{hi} h_{t-1} + b_i) \tag{5}$$

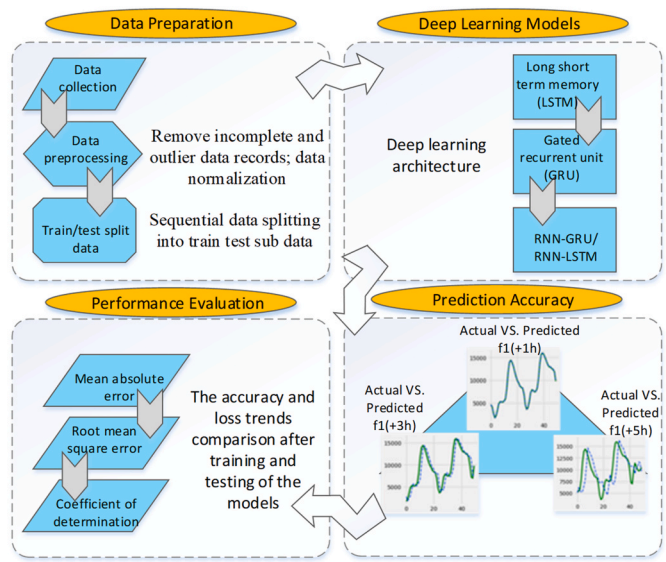


Fig. 7. Flowchart diagram of the proposed scheme described in four blocks.

$$o_t = \sigma(w_o x_t + w_{ho} h_{t-1} + b_o) \tag{6}$$

$$c_t = i_t \otimes (\psi(w_c x_t + w_{hc} h_{t-1} + b_c)) + c_{t-1} \otimes f_t \tag{7}$$

$$h_t = \sigma(o_t \otimes \psi c_t) \tag{8}$$

where, $x_t \in \Re^M \Re$ is a dataset, and M denotes the dimensions (i.e., number of input features) at a time step t. Also, $h_{t-1} \in \Re^N$ with exponent N represents the number of neurons in the LSTM layer. The termsf_t, i_t and o_t symbolize the forget, input, and output gates, respectively. The term c_t is the output of the input modulation and cell unit and h_t is the final output from the LSTM. w_f, w_i, w_o, are the weight vectors applied at the forget gate, input gate, and output gate, and memory cell state, respectively. w_{hf}, w_{hi}, w_{ho}, w_{hc}, are the weight vectors applied to the recurrent input. b_f, b_i, b_o, b_c are the biases and $\otimes$ refer to point wise multiplication. The activation functions tanh (ψ) and sigmoid (σ) for the output of LSTM are constrained to the ranges of $(-1,\ 1)$ and $(0,\ 1)$, respectively.

For many applications, the prediction performances of GRU and LSTM models are similar, although LSTM is computationally more time-consuming. That additional computation time enables LSTM to create deeper networks that can exploit more complex relationships between the variables spanning more time points.

The vanishing and exploding gradient problems in machine learning models can cause consequential variations from one iteration to the next as the model weights changed by large magnitudes leading to model instability and inefficient learning. The vanishing gradient problem can also impact RNN models. It refers to the inefficient transfer of gradient information from the model output to update the weights applied to the layers at the input end for subsequent model iterations. Careful selection of activation functions based on trial and error is required to overcome this problem in GRU and LSTM models.

2.1.3. Support vector regression

In addition to DL techniques to forecast temporal wind power (Section 2.1.2), the same datasets are evaluated, for comparative purposes, using an effective machine learning regression model. The support vector regression (SVR) model is used here to perform this task. SVR uses the same principle as a support vector machine (SVM), but for the regression problems, which have been successfully used for short-term time series prediction of wind power [50]. SVR functions by decreasing the prediction margin error by employing kernel functions.

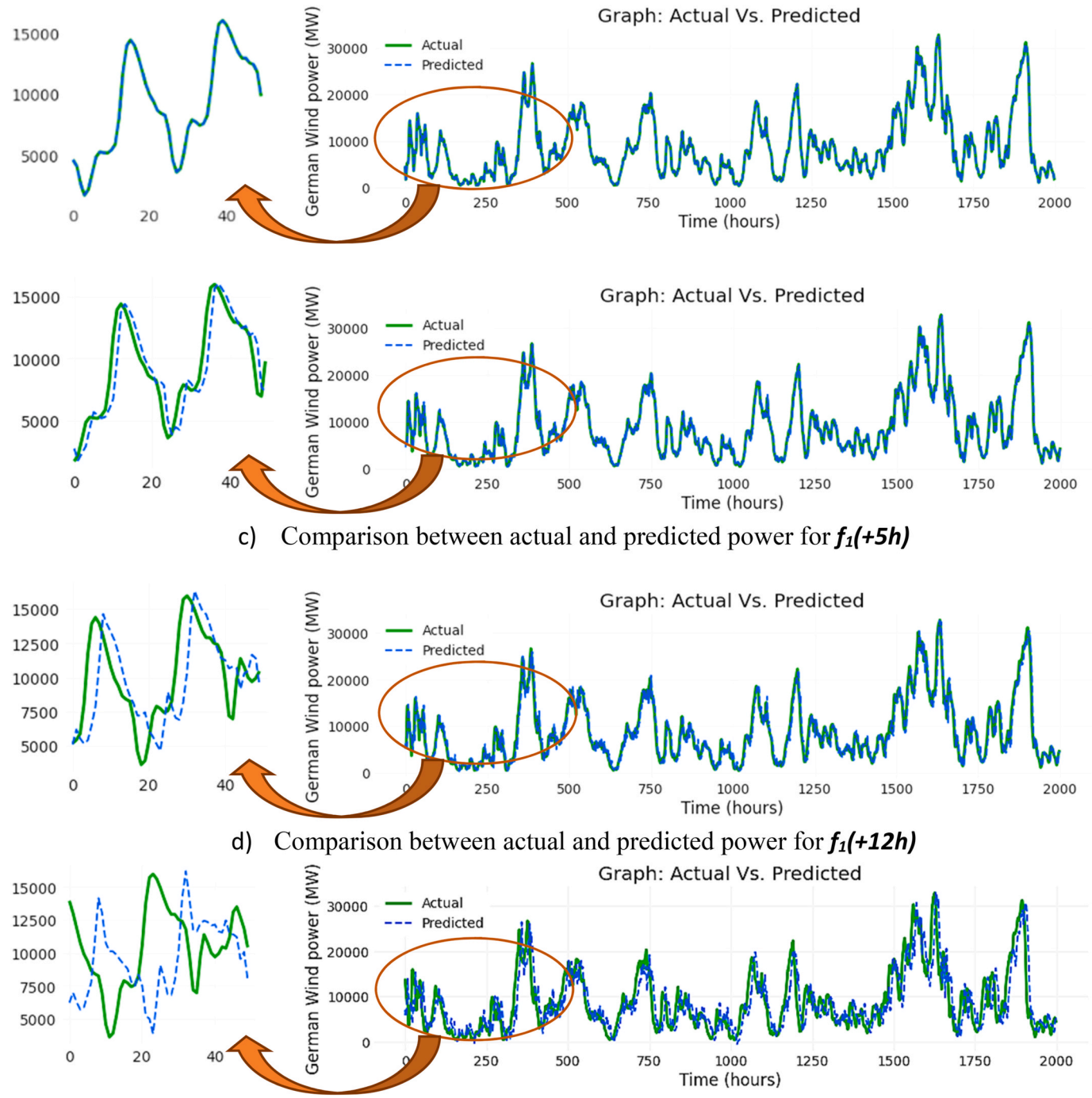


Fig. 8. Comparison of actual versus predicted wind power prediction (MW) by the RNN-GRU model with f_1 *(t)* for hours ahead +1 h, +3 h, +5 h, and +12 h. (Total 8784 = hours in a leap year identifier is used) from these 6148 h data is employed for training and the remaining 2636 h are used for testing (from the testing set, the MW values for the first 2000 h are displayed in this figure).

SVR prediction outcomes can be enhanced by tuning its control parameters. Heuristic and grid search techniques can be usefully employed to find the optimal SVR control parameter values [51]. The SVR algorithm essentially adopts a nonlinear map of the actual (measured) data transforming it into a high dimensional feature space. SVR is a non-parametric algorithm and it can be configured to use a linear kernel to find a linear vector that optimally separates the dataset into groups (or classes for SVM) in multidimensional feature space. SVR models are frequently compared to multi-layer perceptron (MLP), and other artificial neural networks (ANN) architectures in terms of their prediction performance. As the number of features of a dataset approaches the number of data records the performance of SVR models deteriorates. Generally, SVR models require more extensive feature selection to outperform MLP models. Both SVR and MLP are prone to over fitting datasets and rigorous independent testing is required to identify the extent, if any, of over fitting associated with specific datasets.

The basic mathematical formula employed by SVR is expressed as equation (9).

$$y = f(x) = \sum_{i=1}^{m} w_i x_i + b \tag{9}$$

where x_i denotes the input variable values, w_i are weights applied to the input, b represents an applied bias factor, y is the dependent variable

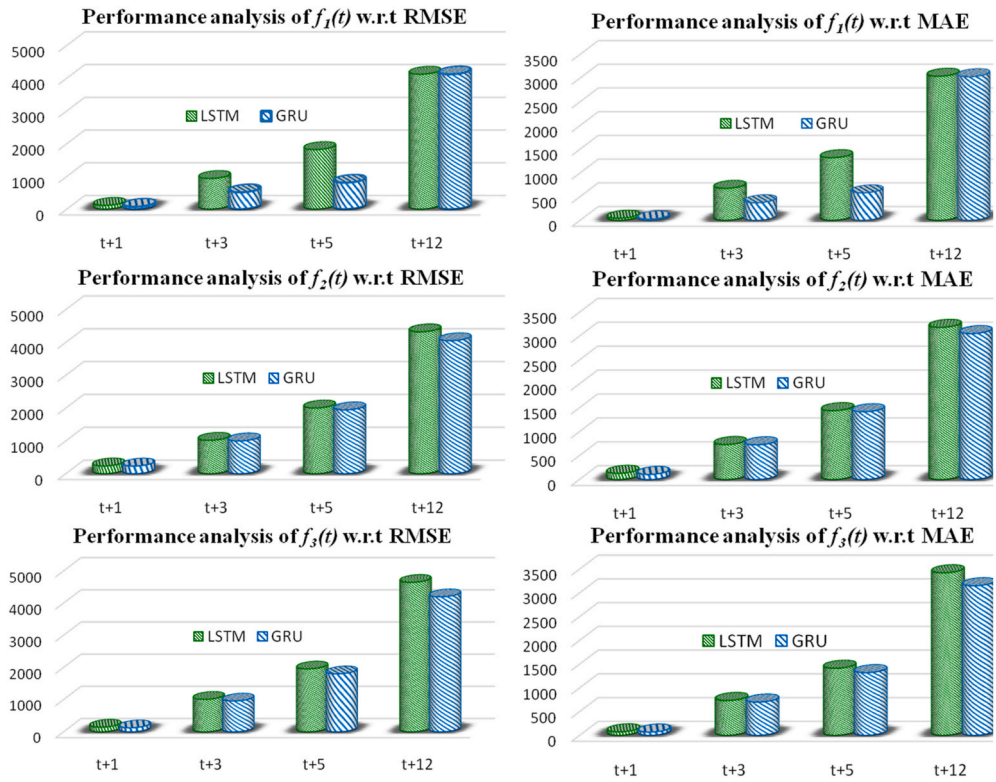


Fig. 9. Comparison of MAE and RMSE for German country-wide 2016 wind power MW predictions at various hours ahead for the RNN-GRU and RNN-LSTM models applied to the complete dataset for Case 2 (70-30% data division).

value (in this study wind power expressed in megawatts, MW) being predicted, and m is the number of data records in the dataset.

SVR models can effectively handle many non-linear datasets. A range of kernel functions can be used to adapt SVR and provide it flexibility beyond its capabilities with a basic linear kernel function. For instance, the radial basis function kernel (RBF) [52,53]. The RBF is generally better able to handle non-linear relationships between dataset features than the linear kernel. Equations (10) and (11) define the linear kernel function used to map data into higher dimensional feature space (i.e., kernel space) to achieve optimal prediction of the dependent variable.

$$f(x) = \sum_{i=1}^{m} (\alpha_i - \alpha_i^*) k(x_i - x) + b \tag{10}$$

$$k(x_i - x) = \varphi(x_i) \cdot \varphi(x) \tag{11}$$

where, $k(x_i - x)$ is the formula component defines the kernel function (equation (10)), and $\varphi(x)$ denotes the input features in the kernel (equation (11)).

The RBF kernel function is expressed mathematically in equation (12).

$$\sigma : k(x_i, x) = \exp\left(-\|(x_i - x)\|^2 / 2\sigma^2 \right) \tag{12}$$

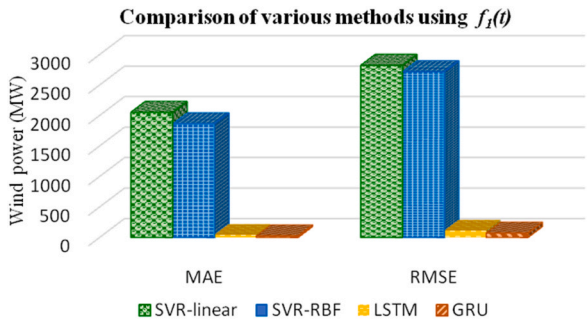
where, σ is the RBF kernel's control parameter that requires tuning.

The two configurations of SVR (linear kernel and RBF kernel) are used in this study to compare their short-term predictions for the non-linear wind power time series considered with predictions made by the two deep learning RNN models developed.

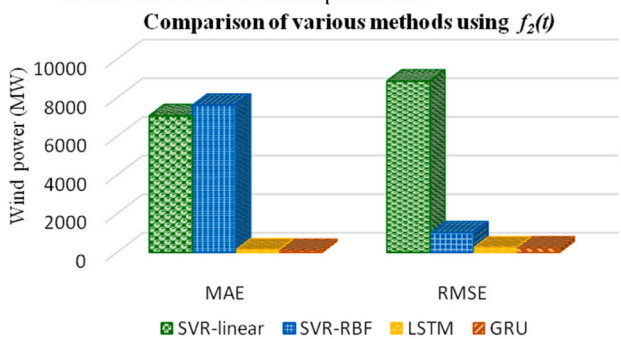
2.2. Data preprocessing

The complete dataset (8784 hourly data records) is initially divided as follows:

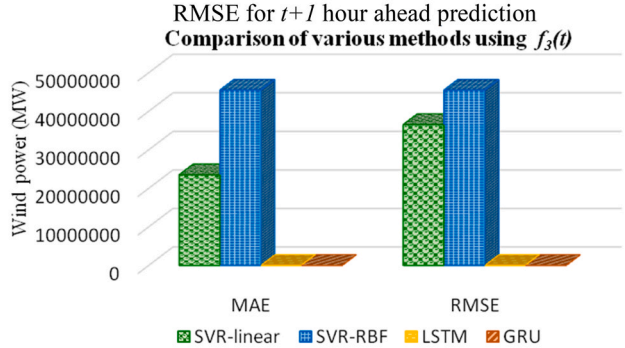
1) *Division into training and testing subsets.* A model's network configurations are adjusted in terms of weights and biases using the training subset. Once the network configurations are optimized taking into consideration all the data records available in the training subset, the optimum fit or configuration is applied to the testing subset, which has been held independently during the training process. The model complexity of the proposed model is evaluated through model size and data complexity. As, we deal with the multivariate time series datasets commonly display some form of temporal dependencies, i.e., periodicity and/or seasonality. Consequently, it is not appropriate to arbitrarily or randomly shuffle the data records, when dividing them into training and testing subsets. In this study, a trial and error approach is applied to evaluate the prediction performance of four different data divisions into training and testing subsets. These divisions are: case 1 (80-20%); case 2 (70-30%); case 3 (60-40%); and, case 4 (50-50%), as discussed in section 2.3.
2) *Scaling of features.* The value ranges of some of the variables in the dataset are quite large. The Robust Scalar [46] function is used to simplify the scaling of the features. It does so by applying statistics that render the dataset robust to outliers. This avoids biases and certain variables and/or outlying data records having undue influence on the algorithms. Function scaling via normalization assists in the rapid convergence of the gradient descent algorithms used by the RNN models. The 2016 wind power hourly dataset evaluated includes several large values for certain variables that represent outliers from the main variable distributions. The application of a filter using Robust Scalar ensures that the scaling of the data is not unduly affected by these outlying values such filtering is applied to the training and testing data subsets separately. This ensures that there are no biases in the test sample since they are unknown to the trained model during the training process. The robust Scalar function is set up to eliminate the median and then scale the data using the quartile range from the 25th quartile to the 75th quartile. The resulting dataset's scaled value ranges are significantly greater than raw data



a) Comparison bar charts of feature set $f_1(t)$ in terms of MAE, RMSE for $t+1$ hour ahead prediction



b) Comparison bar charts of feature set $f_2(t)$ in terms of MAE, RMSE for $t+1$ hour ahead prediction



c) Comparison bar charts of feature set $f_3(t)$ in terms of MAE, RMSE for $t+1$ hour ahead prediction

Fig. 10. Comparison of MAE and RMSE of wind power (MW) predictions from the SVR (linear), SVR (RBF), LSTM, and GRU models using a feature sets $f_1(t)$, $f_2(t)$, $f_3(t)$ with the Case 2 splits (Table 2).

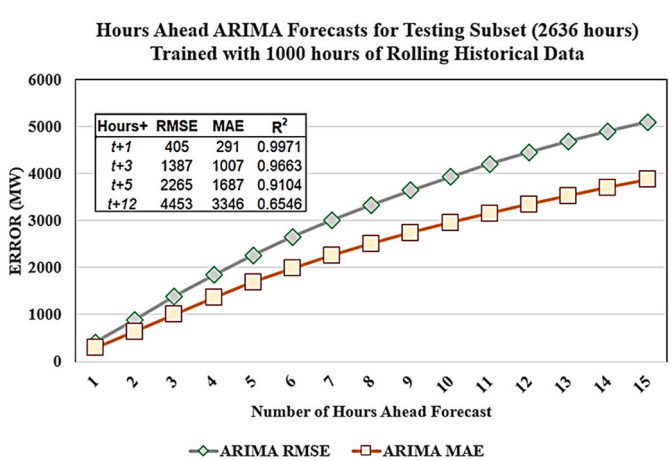


Fig. 11. Results of the ARIMA model for $t+1$ to $t+15$ h ahead.

feature ranges following adjustment with the Robust Scalar. The dataset's scaled feature values are then normalized to each lie within the range [-1, 1].

3) *Selecting data records for short-term forecasting.* Temporal forecasting necessarily requires the consideration of historical information and most of the machine learning models are used 2D array (samples, features) data representation. On the other hand, LSTM can capture the patterns in the time step dimension with a 3D array (samples, time step, and features). Hence, it is necessary to transform the time series segments to be used for forecasting into a temporal regression problem involving specified lag periods (time step). "Lag" (L) refers to the number of time steps (that is hours for the dataset evaluated) to look back over and consider the feature values for use in estimating the next (and/or subsequent) hour(s) into the future. Fig. 4 represents the Lag period used in proposed models. A value of L = 7 (i.e., the seven previous hours of data records), established by the trial and error, was found to be the most suitable lag value for forecasting the wind power dataset considered for the very near future (e.g.$t+1$). Table 1 lists prediction performances for selected lag time assessments. The specified lag period constrains the model to consider data only from those historical lag hours (i.e., *t back to t-7*) are available to the model at the time of forecasting, but no information from $t+n$ is made available to the model.

2.3. Network architectures

The configuration and control parameters applied to the RNN-GRU and RNN-LSTM models developed for this study are listed in Table 2. SVR with the linear kernel is simply implemented without control value options. SVR with the RBF kernel is applied in this study with specified values, established by trial and error, assigned to three key control values: gamma (C) value of 100 (equivalent to the inverse of the radius of influence); a polynomial degree of 3; and, an epsilon value of $1*10^{-7}$ (a margin of tolerance within which no penalty is applied to errors during training) is used.

The control parameters applied to the RNN models are also listed in Table 2. The RNN-GRU/LSTM models (using the Keras deep learning platform) are configured with two layers and adopt the Adam optimizer. The dropout control metric in the GRU/LSTM models is implemented between the layers because it helps to reduce overfitting tendencies. The important hyper-parameters for the proposed algorithms are the number of its hidden neurons, the optimizer, and the number of lags (look back) essential to be decided. Parameter optimization plays a very important role to achieve higher accuracy, better efficiency, and good convergence. We executed the Grad student descent/(babysitting AKA) trial and error rule to optimize each hyper parameter as listed in Table 2, keeping the rest constant, and then selected the best parameter values as a trade-off between accuracy and time consumption. The experimental results for parameter optimization used different optimizers; learning rate and the number of hidden neurons are achieved by computing the best score of the validation set, Based on the score number of lags is selected. All models evaluated are coded in Python and executed on a computer with windows 10 with the following specifications: Intel Core (TM) i7-8550U, 1.80-GHz processor, 16 GB of DDR4 RAM, and 1 TB of the hard disk.

Four distinct cases with different training/testing subset splits (Table 3) are evaluated with each model. The primary objective with each case is for each model to predict the wind power magnitude for the next 1 h, 3 h, 5 h, and 12 h, applying the model in a forecasting configuration.

2.4. Prediction accuracy measures used to evaluate model performance

The root mean squared error (RMSE) expressed in power units (MW) is the statistical measure of prediction accuracy (equation (13)) used to assess the performance of each model.

Table 6
MW prediction performance for four models evaluated for all three feature sets to predict the $t+1$ h wind power. Each evaluation predicts the values for 1756 data records (Case 1, Table 3).

Models	+1 h-ahead for f_1 (t)		+1 h-ahead for f_2 (t)		+1 h-ahead for f_3 (t)	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
SVR-linear	6400.50	6606.78	9230.46	11480.04	11528.97	13116.45
SVR-RBF	5360.47	6136.98	9031.01	11358.38	11463.75	12908.93
RNN-LSTM	45.06	95.73	155.37	244.62	108.90	209.84
RNN-GRU	**44.017**	**97.69**	**95.07**	**186.41**	**97.71**	**187.61**

Table 7
MW prediction performance for four models evaluated for all three feature sets to predict the $t+1$ h wind power. Each evaluation predicts the values for 2636 data records (Case 2, Table 3).

Models	+1 h-ahead for f_1 (t)		+1 h-ahead for f_2 (t)		+1 h-ahead for f_3 (t)	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
SVR-linear	6242.30	6370.21	8694.83	10805.76	9949.56	11223.11
SVR-RBF	6515.31	8304.31	8501.04	10672.37	9616.86	10757.56
RNN-LSTM	43.23	112.53	129.87	234.45	71.11	145.63
RNN-GRU	**26.45**	**73.01**	**106.76**	**231.55**	**67.24**	**125.08**

$$RMSE = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_{(actual)} - y_{(predcited)}\right)^2} \tag{13}$$

where $y_{(actual)}$ is the actual or measured value of wind power (MW), $y_{(predcited)}$ is the forecast or predicted value output by the model and N is the number of data records evaluated.

The mean absolute error (MAE) is a useful additional statistical measure of prediction accuracy (equation (14)) recorded. It focuses on absolute error rather than the squared error involved in the RMSE metric.

$$MAE = \frac{1}{N}\sum_{i=1}^{N}\left|y_{(actual)} - y_{(predcited)}\right| \tag{14}$$

Another worthwhile prediction accuracy performance indicator considered is the coefficient of determination (R^2) as expressed by equation (15).

$$R^2 = 1 - \frac{\sum_i\left(y_{(actual)} - y_{(predcited)}\right)^2}{\sum_i\left(y_{(actual)} - y_{(\mathrm{mean})}\right)^2} \tag{15}$$

3. Dataset evaluated

3.1. Country-wide wind power dataset for Germany covering year 2016

The dataset evaluated in this study is taken from the Open Power System Data (OPSD) and is comprised of hourly-averaged German data records [27]. This one year of data involves 8784 hourly records (sampling all 366 days of 2016). A summary of the variable values for Wood's compiled dataset is shown in Table 4.

The dataset includes a broad set of independent variables including wind speeds, other atmospheric and environmental conditions, and the power price index. Three wind speed metrics are V1, V10, and V50 (2 m above displacement height), h2 (10 m above displacement height), and h3 (50 m above ground), respectively. An environmental indicator Z0 refers to surface roughness length [m]. The temperature component T in degrees centigrade is measured at 2 m above the displacement height. Other metrics related to atmospheric conditions are air density [kg/m3] at the ground level (Rho), and air pressure (Pa) at the ground level. The power price index variable is one devised by Wood [27] computed day-ahead power price in Euro/MW (Pr-Annual Mean Pr)/(Annual Mean Pr). For further analysis, all independent variable values have been scaled and then normalized [−1, 1] using the Robust Scaler function (see Section 2.2). Fig. 5 displays the value distributions for each input variable.

Each variable value in each data record of the country-wide dataset is averaged for each hourly period from data recorded across the whole country. This means that the variable values in each hourly data record are compiled in the source OPSD dataset from measurements recorded at many thousands of individual wind farms. This also means that the hour-to-hour MW variations displayed by the compiled data records reflect both changes in the prevailing meteorological conditions and changes in the collective availability/contribution from all wind power sources.

3.2. Feature selection for input variable reduction

Hourly measurements for each input variable associated with each wind power data record are evaluated to predict the power values for each of the 8784 h in 2016. Fig. 6 displays the characteristics of the

Table 9
MW prediction performance for four models evaluated for all three feature sets to predict the $t+1$ h wind power. Each evaluation predicts the values for 4392 data records (Case 4, Table 2).

Models	+1 h-ahead for f_1 (t)		+1 h-ahead for f_2 (t)		+1 h-ahead for f_3 (t)	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
SVR-linear	5267.59	5326.05	7566.50	9535.26	9241.84	10486.28
SVR-RBF	6583.89	8569.71	7371.46	9390.73	9024.21	10120.59
RNN-LSTM	42.21	84.08	154.49	221.11	161.02	261.38
RNN-GRU	**24.23**	**53.08**	**95.44**	**195.91**	**127.61**	**203.20**

Table 8
MW prediction performance for four models evaluated for all three feature sets to predict the $t+1$ h wind power. Each evaluation predicts the values for 5270 data records (Case 3, Table 3).

Models	+1 h-ahead for f_1 (t)		+1 h-ahead for f_2 (t)		+1 h-ahead for f_3 (t)	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
SVR-linear	7791.59	9856.10	9890.17	11082.44	9949.56	11223.11
SVR-RBF	7844.96	9981.54	10226.54	11427.89	9616.86	10757.56
RNN-LSTM	40.83	86.72	126.37	215.99	88.22	167.24
RNN-GRU	**30.40**	**55.61**	**99.03**	**202.81**	**76.75**	**153.60**

Table 10
Statistical results of a paired *t*-test comparing the forecasting accuracies of the RNN-GRU and RNN-LSTM models applied to the dataset configured for Case 4, feature set f_3 *(t)* and hours ahead $t + 1$.

Paired Samples Statistics

	Mean	N	Std. Deviation	Std. Error Mean
GRU_MAE	118.80	40	8.093	1.280
LSTM_MAE	133.13	40	9.466	1.497

a)

Paired Samples Correlations

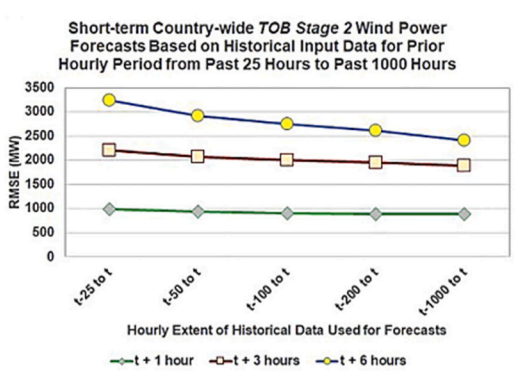
	N	Correlation	Sig.
GRU_MAE & LSTM_MAE	40	-.187	.248

b)

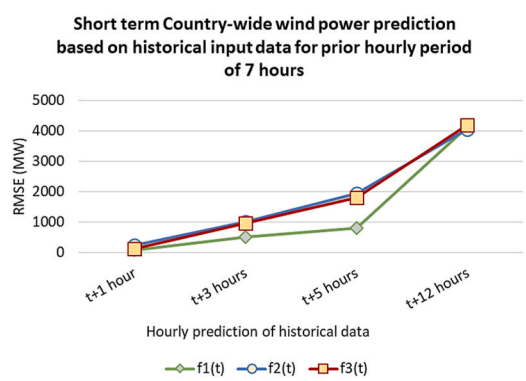
Paired Samples Test

	Paired Differences					t	df	Sig. (2-tailed)
	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval of the Difference				
				Lower	Upper			
GRU_MAE - LSTM_MAE	14.337	13.554	2.143	-18.672	-10.002	-6.690	39	.000

c)



a. Short-term forecasting accuracy compared for $t + 1$, $t + 3$, and $t + 6$ h ahead using five different ranges of historical data periods to provide the TOB Stage 2 forecasts optimized with the best solution derived from prediction analysis, Wood [27]



b. Short-term forecasts compared for $t+1$, $t+3$, $t+5$, and $t+12$ hour ahead using a static range of previous information by employing the three different feature sets with the RNN-GRU model for Case 2.

Fig. 12. Short-term forecasting accuracy comparisons of the RNN-GRU model with the previously published work on the same country-wide wind power dataset.

correlation relationship between the eight input variables (V1, V10, V50, Z0, T, Rho, Pa, PrI) and the dependent variable (wind power, MW) expressed in the form of a heat map.

The heat map (Fig. 6) is a useful tool for visualizing correlation relationships among variables, with dark colors representing strong correlations and light colors representing weak and/or negative correlations. Among the 8 input variables (features), those showing the strongest correlations with MW are the wind speed variables (V1, V10, and V50), and these are designated as the f_1 (*t*) feature set. The remaining five input variables (Z0, Rho, Pa, PrI, and T) are designated as the f_2 (*t*) feature set. A feature set $f_3(t)$, including all eight input features (V1, V10, V50, Z0, Rho, Pa, PrI, and T), is also specified as the benchmark feature set for the prediction models.

4. Results

Following data preprocessing, the 8784 data records of the German country-wide 2016 wind power dataset, each consist of 8 input variables/features. Feature reduction is carried out based on the input variable's potential influence on the MW dependent variable. That

influence is established based on the correlation coefficient heat map (Fig. 6), which identifies specifically three highly relevant features (wind velocities V1, V10, and V50). This analysis led to the establishment of three distinct feature sets, $f_1(t)$, $f_2(t)$, and $f_3(t)$, as discussed in section 3.2. The four cases (different training/testing subset splits) described in Table 3 are evaluated using each of these three feature sets. They are configured in separate evaluations to predict MW at specific hours ahead: $t+1$ (1 h ahead), $t+3$ (3 h ahead), $t+5$ (5 h ahead), and $t+12$ (12 h ahead). Table 5 describes the MW prediction accuracy for those evaluations, revealing that the RNN-GRU models outperform the RNN-LSTM models in all cases evaluated. Fig. 7 illustrates the proposed schema.

The results displayed in Fig. 8 are for the feature set ($f_1(t)$) evaluated with the RNN-GRU model trained and configured to predict wind power for the data records in the German 2016 country-wide hourly compiled dataset [25]a certain number of hours forward (i.e., +1 h, +3 h, +5 h, and +12 h in Fig. 7 (a, b, c, and d, respectively). Fig. 8 plots actual versus predicted MW values for the hours 0 to 2000 of the testing subset, i.e., the first 2000 h of the period starting September 13, 2016; 04:00:00 and ending 5/12/2016; 11:00:00). There is good agreement between the actual and predicted values in all four forecasts but the prediction accuracy for the +1 h ahead forecast is much better than for the +12 h ahead prediction. Fig. 9 verifies the superiority of the RNN-GRU model compared to the RNN-LSTM model in terms of the MAE and RMSE metrics for the $t+1$, $t+3$, $t+5$, and $t+12$ h ahead wind power predictions. This is the case for all three feature sets ($f_1(t)$, $f_2(t)$, $f_3(t)$). It is also apparent from Fig. 9 that the prediction accuracy deteriorates substantially beyond $t+5$ h ahead.

Fig. 10(a–c) displays comparative results for the test data records for Case 2 (2635 records) for all three features evaluated by the four models: SVR (linear); SVR (RBF); RNN-LSTM; and, RNN-GRU. It is apparent that the two RNN models substantially outperform the SVR linear and RBF models in forecasting wind power using this dataset. Moreover, $f_1(t)$, achieves the highest forecasting accuracy among all models.

The outcomes of an Autoregressive Integrated Moving Average (ARIMA) model, implemented on the 30% testing subset, were used to assess the detailed prediction performances of the RNN and SVR models are presented in Fig. 11. The ARIMA model, coded in Python, uses only the MW time series as input. Each of the 2636 data records predicted by the ARIMA model is based on training using the preceding 1000 h of data. Trial and error assessment established the optimum ARIMA model control values are P = 2; D = 0; Q = 2, where P represents lag, D seasonality adjustments, and Q the moving average window. For its $t+1$ h ahead forecast, the ARIMA trained model achieved an RMSE prediction accuracy of 405 MW, but its accuracy deteriorated rapidly for forecasting additional future hours (Fig. 11). The prediction accuracy achieved by the ARIMA model is substantially inferior to that achieved by the RNN models (Table 5) although, superior to that achieved by the SVR models (Fig. 10).

Table 6–9 present the forecasting accuracy, in terms of MAE and RMSE, by all four RNN and SVR models (with settings as specified in Table 2) for various splits of the data records into training and testing subsets; Cases 1 to 4 represents split 80-20%, 70-30%, 60, 40%, and 50, 50%.

Each table represents the results of three feature subsets ($f_1(t)$, $f_2(t)$, and $f_3(t)$) for 1 h ahead $(t+1)$ wind power forecasting. Considering the results collectively from Tables 6–9, the RNN-GRU model revealed better performance for short-term wind power forecasting for all cases. These results also indicate that different training and testing subset distributions (i.e. an increase/decrease of unseen examples) do not lead to substantial variations in the forecasting performance of each model. The best model performances are, however, for Case 4 (Table 9) because they involve the largest number of data records assigned to the training subset.

5. Statistical validation demonstrating the superiority of the RNN-GRU model

The proposed variants of RNN models are evaluated for a more comprehensive and quantitative understanding of the improved performance by running them for forty independent executions, and the predicted errors are visualized in terms of paired sample correlation statistics and paired sample correlation as shown in Table 10. For this purpose, a paired *t*-test is conducted using the testing subset (50-50% data split) of Case 4, feature set $f_3(t)$, and hours ahead $t+1$. The paired *t*-test is applied to two sets of evaluation results performed on the sample set of data records to establish the 95% confidence interval regarding the difference between the means of the evaluation sets of multiple iterations of the two models. The error metric used for the paired *t*-test is the MAE values for the RNN-GRU and RNN-LSTM models (Table 10). Table 10(a) identifies a significant difference between the MAE values of the GRU (M = 118.80, SD = 8.093) and LSTM (M = 133.13, SD = 9.466) models. Table 10(b) shows the correlation coefficient between the two sets of a model in which evaluations are negative and that they are distinct from each other with a significance level of 0.248. The mean and standard deviations of the differences in MAE between the GRU and LSTM are −14.337 and 13.554, respectively (Table 10c). The 95% confidence interval of that difference is expressed as a lower bound of −18.672 and an upper bound of −10.002 (Table 10c). The paired *t*-test results are t = −6.690 and p = 0.000. The value of significance for the two-tailed test is less than 0.05, confirming that the differences between the evaluation sets are statistically significant and that there is no significant correlation between the prediction values of the GRU and LSTM models and therefore, depicts the significance of our proposed model, RNN-GRU.

6. Discussion

Wind speed is affected by a variety of elements, including temperature, humidity, surface obstructions, and air pressure, resulting in stochastic and nonlinear behavior. Conventional machine learning models frequently fail to accurately forecast complicated sequences based on multiple influencing variables. Power systems, on the other hand, need high-precision wind power predictions due to higher-precision predictions can save more power system operating costs. As a result, it is desirable to employ more sophisticated models to generate greater accuracy forecasts. When compared to existing prediction models SVR, ARIMA, and RNN-LSTM, the suggested RNN-GRU technique is more computationally efficient and substantially more accurate for the $t+1$ h ahead forecasts but becomes progressively less accurate for the $t+3$, $t+5$, and $t+12$ h ahead forecasts. The error rates of each model grow as the number of prediction steps increases. This is most likely due to the accumulation of forecast errors in the recursive forecasting process.

As the experimental results in Fig. 9 have demonstrated that the more step ahead prediction gives the higher error rate. The 3-, 5-, and 12-steps in terms of bar charts are not the time window of the prediction results, but the number of steps ahead prediction. The purpose of RNN-GRU is for multi-step forecasting and the use of multi-location feature learning to complete the forecasting tasks of country-wide wind farms in Germany. For each wind power prediction task, 1, 3, 5, and 12 output neurons are required to complete the multi-step prediction in 1-, 3-, 5-, and 12-steps, respectively, i.e., the number of steps ahead prediction is equal to the number of output neurons. RNN-GRU has more stable gradients and is easier to train since its back-propagation path is distinctive from the RNN-LSTM, minimizing the effects of multiple gates.

For further evaluation of the RNN-GRU model, the short-term forecasting results are compared with previous work conducted on the same dataset by Wood [27] applying the transparent open box (TOB) optimized data matching algorithm to conduct short-term hourly predictions. The optimum TOB configuration used for MW predictions was

established by tuning the algorithm over the full-time series (Steps 1–14) and then applying that tuned model in forecast mode (Step 15). That tuned model provided TOB Stage 2 short-term forecasts for three different hours ahead ($t+1$, $t+3$, $t+6$) spread sequentially across the full year. Fig. 12 shows that the RMSE error achieved by the TOB model at the $t+1$, and $t+3$ time steps are higher than that achieved by the RNN-GRU model, suggesting that the RNN-GRU model can provide a better short-term forecast up to at least t+5 than the TOB model when applied to this dataset.

The results suggest that the ability of the RNN-GRU model to memorize near-past data over a 7-h lag period is more effective in forecasting up to 5-h ahead than data matching applied over a long historic period.

7. Conclusions

Two novel recurrent neural network – gated recurrent unit (RNN-GRU) and -long short term memory (RNN-LSTM) deep learning models were developed in this study to deliver multi-time-step (hours) ahead short-term forecasting of wind power. Both models demonstrated computational efficiency with good learning and generalization capabilities in the nonlinear feature space typical of wind power time-series datasets. Additionally, three feature subsets and four cases were conducted based on the hourly dataset for Germany. Those models generated efficient, reliable, and robust predictions for $t+1$ to $t+12$ h ahead in terms of the statistical accuracy measures mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determinations (R^2). Specifically, the results obtained compared to published data for short-term wind power forecasts for the same dataset. When compared to the machine learning method, support vector regressor (SVR) configured with either linear or RBF kernels, the developed RNN models produce better predictions. Quantitatively, the MAE and RMSE values in MW for one-step (hour $t+1$) prediction of the best performing RNN-GRU model are 44.01, 26.45, 30.40, 24.23, and 97.69, 73.01, 55.61, 53.08, respectively, applied to four different allocations of the data between training and testing subsets. The case 2 split of 70% training/30% testing generated the lowest errors. The RNN-GRU model substantially outperformed RNN-LSTM, SVR-RBF, and SVR-Linear models in its hours-ahead wind power prediction. Moreover, RMSE, and MAE values in MW for case 2 applied to three different feature selections are 26.45 and 73.01 for the feature set f_1 (three wind speed input variables only); 106.76 and 231.55 for f_2 (five input variables excluding wind speed), 67.24 and 125.08 for f_3 (all eight input variables), respectively. The results achieved by the RNN-GRU model suggest that it is worthy of evaluation with other time-series data sets showing high volatility in the dependent variable over short-term cycles, such as those associated with the solar, geothermal, and other renewable power generation projects. The results presented suggest that the RNN-GRU model has the potential to be applied to other time-series data sets showing high volatility in the dependent variable over short-term cycles, such as those associated with the solar, geothermal, and other power generation projects.

CRediT author statement

Farah Shahid: Conceptualization, Methodology, Validation, Investigation, Writing an original draft, Visualization. David A Wood: Data Curation, Writing-Review & Editing, Visualization. Humaira Nisar: Visualization, Project administration. Aneela Zameer: Resources, Project administration. Steffen Eger: Visualization, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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References

[1] Losekann L, Tavares A. Potential for cooperation in the dissemination of renewable energy and natural gas among BRICS countries. 2021.
[2] Sengers B, et al. A physically interpretable statistical wake steering model. Wind Energy Science Discussions; 2021. p. 1–23.
[3] Gwec. https://www.renewableenergyworld.com/wind-power/gwec-offshore-wind-is-estimated-to-surge-to-234-gw-by-2030/#gref 2021.
[4] Rechsteiner R. German energy transition (Energiewende) and what politicians can learn for environmental and climate policy. Clean Technol Environ Policy 2021;23(2):305–42.
[5] Liu X, Lin Z, Feng Z. Short-term offshore wind speed forecast by seasonal ARIMA - a comparison against GRU and LSTM. Energy 2021;227:120492.
[6] Başakın E, et al. A new insight to the wind speed forecasting: robust multi-stage ensemble soft computing approach based on pre-processing uncertainty assessment. Neural Comput Appl 2022:34.
[7] Alkesaiberi A, Harrou F, Sun Y. Efficient wind power prediction using machine learning methods: a comparative study. Energies 2022;15(7):2327.
[8] Yan J, Ouyang T. Advanced wind power prediction based on data-driven error correction. Energy Convers Manag 2019;180:302–11.
[9] Wang K, et al. Wind power short-term forecasting hybrid model based on CEEMD-SE method. Processes 2019;7(11):843.
[10] Wu Q, Lin H. Short-term wind speed forecasting based on hybrid variational mode decomposition and least squares support vector machine optimized by bat algorithm model. Sustainability 2019;11(3):652.
[11] Özen C, Deniz A. Short-term wind speed forecast for Urla wind power plant: a hybrid approach that couples weather research and forecasting model, weather patterns and SCADA data with comprehensive data preprocessing. 0309524X221088612. Wind Engineering; 2022.
[12] Perr-Sauer J, et al. Short-term wind forecasting using statistical models with a fully observable wind flow. In: Journal of physics: conference series. IOP Publishing; 2020.
[13] Dabrowski JJ, et al. Enforcing mean reversion in state space models for prawn pond water quality forecasting. Comput Electron Agric 2020;168:105120.
[14] Aly HH. An intelligent hybrid model of neuro Wavelet, time series and Recurrent Kalman Filter for wind speed forecasting. Sustain Energy Technol Assessments 2020;41:100802.
[15] Zhang D, et al. A novel wind speed forecasting based on hybrid decomposition and online sequential outlier robust extreme learning machine. Energy Convers Manag 2019;180:338–57.
[16] Iannace G, Ciaburro G, Trematerra A. Wind turbine noise prediction using random forest regression. Machines 2019;7(4):69.
[17] Botha N, Walt CMvd. Forecasting wind speed using support vector regression and feature selection. In: 2017 pattern recognition association of South Africa and robotics and mechatronics (PRASA-RobMech); 2017.
[18] !!! Invalid citation !!! .
[19] Yu Y, et al. Probabilistic prediction of regional wind power based on spatiotemporal quantile regression. IEEE Trans Ind Appl 2020;56(6):6117–27.
[20] Gonçalves C, et al. Forecasting conditional extreme quantiles for wind energy. Elec Power Syst Res 2021;190:106636.
[21] Liu Y, et al. Wind power short-term prediction based on LSTM and discrete wavelet transform. Appl Sci 2019;9(6):1108.
[22] Shahid F, et al. A novel wavenets long short term memory paradigm for wind power prediction. S0306261920306103 Appl Energy 2020;269(C).
[23] Chai SS, et al. Radial basis function (RBF) neural network: effect of hidden neuron number, training data size, and input variables on rainfall intensity forecasting. Int J Adv Sci Eng Inf Technol 2019;9(6):1921–6.
[24] Priya CB, Arulanand N. Univariate and multivariate models for Short-term wind speed forecasting. Mater Today Proc 2021.
[25] Yaseen ZM, et al. An enhanced extreme learning machine model for river flow forecasting: state-of-the-art, practical applications in water resource engineering area and future research direction. J Hydrol 2019;569:387–408.
[26] Lytras MD, Chui KT. The recent development of artificial intelligence for smart and sustainable energy systems and applications. Multidisciplinary Digital Publishing Institute; 2019. p. 3108.
[27] Wood DA. Country-wide German hourly wind power dataset mined to provide insight to predictions and forecasts with optimized data-matching machine learning. Renew Energy Focus 2020;34:69–90.

[28] Yuan X, et al. Wind power prediction using hybrid autoregressive fractionally integrated moving average and least square support vector machine. Energy 2017; 129.
[29] Tian Z, Li H, Li F. A combination forecasting model of wind speed based on decomposition. Energy Rep 2021;7:1217–33.
[30] Harrou F, Ahmed S, Sun Y. Wind power prediction using bootstrap aggregating trees approach to enabling sustainable wind power integration in a smart grid. Energy Conversion and Management; 2019. p. 201.
[31] Zuluaga C, Álvarez M, Giraldo E. Short-term wind speed prediction based on robust Kalman filtering: an experimental comparison. Appl Energy 2015;156:321–30.
[32] Hu J, Wang J. Short-term wind speed prediction using empirical wavelet transform and Gaussian process regression. Energy 2015;93:1456–66.
[33] Li B-J, et al. Monthly runoff forecasting using variational mode decomposition coupled with grey wolf optimizer based long short-term memory neural networks. 2021.
[34] Shahid F, et al. Wind power prediction using a three stage genetic ensemble and auxiliary predictor. Appl Soft Comput 2020;90:106151.
[35] Sloughter J, Gneiting T, Raftery A. Probabilistic wind speed forecasting using ensembles and bayesian model averaging. J Am Stat Assoc 2010;105:25–35.
[36] Zameer A, et al. Intelligent and robust prediction of short term wind power using genetic programming based ensemble of neural networks. Energy Conversion and Management; 2017. p. 134.
[37] Qu Z, et al. A hybrid model based on ensemble empirical mode decomposition and fruit fly optimization algorithm for wind speed forecasting. Adv Meteorol 2016; 2016:3768242.
[38] Wu J, Zhou T, Li T. A hybrid approach integrating multiple ICEEMDANs, WOA, and RVFL networks for economic and financial time series forecasting. Complexity 2020;2020:9318308.
[39] Lin Z, Liu X. Wind power forecasting of an offshore wind turbine based on high-frequency SCADA data and deep learning neural network. Energy 2020;201: 117693.
[40] Fan H, et al. M2GSNet: multi-modal multi-task graph spatiotemporal network for ultra-short-term wind farm cluster power prediction. Appl Sci 2020;10(21):7915.
[41] Dong W, et al. Short-term regional wind power forecasting for small datasets with input data correction, hybrid neural network, and error analysis. Energy Rep 2021; 7:7675–92.
[42] Liu B, et al. A novel deep learning approach for wind power forecasting based on WD-LSTM model. Energies 2020;13(18):4964.
[43] Altan A, Karasu S, Zio E. A new hybrid model for wind speed forecasting combining long short-term memory neural network, decomposition methods and grey wolf optimizer. Appl Soft Comput 2021;100:106996.
[44] Karasu S, Altan A. Crude oil time series prediction model based on LSTM network with chaotic Henry gas solubility optimization. Energy 2022;242:122964.
[45] Niu Z, et al. Wind power forecasting using attention-based gated recurrent unit network. Energy 2020;196:117081.
[46] Cali U, Sharma V. Short-term wind power forecasting using long-short term memory based recurrent neural network model and variable selection. Int J Smart Grid Clean Energy 2019;8(2):103–10.
[47] Ding M, et al. A gated recurrent unit neural networks based wind speed error correction model for short-term wind power forecasting. Neurocomputing 2019: 365.
[48] Kisvari A, Lin Z, Liu X. Wind power forecasting – a data-driven method along with gated recurrent neural network. Renew Energy 2021;163:1895–909.
[49] Schaefer AM, Udluft S, Zimmermann H-G. Learning long-term dependencies with recurrent neural networks. Neurocomputing 2008;71(13–15):2481–8.
[50] Zendehboudi A, Baseer MA, Saidur R. Application of support vector machine models for forecasting solar and wind energy resources: a review. J Clean Prod 2018;199:272–85.
[51] Kong X, et al. Wind speed prediction using reduced support vector machines with feature selection. Neurocomputing 2015;169:449–56.
[52] Karamichailidou D, Kaloutsa V, Alexandridis A. Wind turbine power curve modeling using radial basis function neural networks and tabu search. Renew Energy 2021;163:2137–52.
[53] Guo Y-m, et al. Time series prediction method based on LS-SVR with modified Gaussian RBF. In: ICONIP; 2012.